Band-1 Hadamard Order 4868 — Miyamoto (Double Paley-I seed H_1216); Miyamoto-only Bonus (NOT in SPK); Gap 2052 Still OPEN

H_608/H_1216/C_1218/H_4868 PASS; 4868 Miyamoto-only bonus (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_4804_Miyamoto`; `2026-09-15_Band1_Hadamard_4516_Miyamoto` (Double Paley-I pattern).
Lab. `/workspace/experiments/band1/hadamard_4868_miyamoto/`.
Cutoff. 2026-09-15T17:48:29Z (~1:48PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 608, 1216 and 4868 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Double(Paley-I) seed H_{1216} (H_{608} = Paley-I($q = 607$); H_{1216} = Sylvester-double(H_{608})). Paley-II path for H_{1216} is **FAIL** ($q = 607 \equiv 3 \pmod 4$; needs $\equiv 1$). Paley-I direct for H_{1216} is **SKIP** ($q = 1215 = 3^5 \cdot 5$ not prime). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 4868 is a **Miyamoto-only bonus** (NOT in SPK): Paley-I FAIL ($q = 4867 = 31 \cdot 157$), Paley-II FAIL ($q = 2433 = 3 \cdot 811$ not prime power), Kronecker FAIL (4×1217). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 4804 on box. This increment ships **order 4868** ($4868 = 4 \cdot 1217$, 1217 prime $\equiv 1 \pmod 4$) via Miyamoto with a **Double(Paley-I) seed** H_{1216}. Symmetric Paley conference C_{1218} over prime 1217; $H_{4868} = \text{Miyamoto}(H_{1216}, C_{1218})$. Unit tests **52/52 PASS**. SPK next gap remains **2052** (OPEN). Order 4868 is **Miyamoto-only** (NOT in SPK). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 4868; Miyamoto; Paley type I; Double Paley-I; Paley conference; constructive gap 2052; SPK; Band-1; Miyamoto-only bonus; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{608} constructed and verified (Paley type I, $q = 607$). 2. H_{1216} constructed and verified (Sylvester double of H_{608}). Paley-II path for H_{1216} **FAIL** ($q = 607 \equiv 3 \pmod 4$). Paley-I direct **SKIP** ($q = 1215 = 3^5 \cdot 5$). 3. C_{1218} constructed and verified (Paley conference, $q = 1217$): $CC^\top = 1217I$, zero diagonal. 4. H_{4868} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 5. SPK: $4868 \notin$ SPK (Miyamoto-only bonus). Paley-I / Paley-II / Kronecker paths for 4868 **rejected**. Next constructive gap remains **2052** (OPEN). Verified H_{608}/H_{1216} enter SPK as Double Paley-I seeds. Prior 4804 remains in SPK via Paley-II prime power 7^4.

2. Arithmetic and eligibility

Quantity	Value
Target order	$4868 = 4 \cdot 1217$
$q = 1217$	prime, $1217 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{1216}
Needed conference	order $1218 = q + 1$
H_{1216} path (used)	Double(Paley-I): $H_{608} = \text{Paley-I}(q = 607)$; $H_{1216} = \text{Sylvester-double}(H_{608})$
H_{1216} Paley-II	**FAIL**: $2(q+1) = 1216 \Rightarrow q = 607 \equiv 3 \pmod 4$ (needs $\equiv 1$)
H_{1216} Paley-I direct	**SKIP**: $q = 1215 = 3^5 \cdot 5$ not prime
Conference path	Symmetric Paley over prime 1217
Paley-I for 4868	**REJECTED**: $q' = 4867 = 31 \cdot 157$ not prime
Paley-II for 4868	**REJECTED**: $q' = 2433 = 3 \cdot 811$ not prime power

Quantity	Value
Kronecker for 4868	**REJECTED**: only 4×1217; $1217 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{608}	Paley type I, $q = 607$	0	0.3276	**PASS**
H_{1216}	Sylvester double of H_{608}	0	0.8308	**PASS**
H_{1216} Paley-II	$q = 607 \equiv 3 \pmod 4$	—	—	**FAIL** (needs $\equiv 1$)
H_{1216} Paley-I direct	$q = 1215 = 3^5 \cdot 5$	—	—	**SKIP** (not prime)
C_{1218}	Paley conference, $q = 1217$	0	1.4666	**PASS**
H_{4868}	Miyamoto(H_{1216}, C_{1218})	0	51.0176	**PASS**

Primary pipeline wall (incl. unit tests + SPK refresh) ≈ 55.1211 s. Compressed ± 1 matrices saved as `H608_pm1.npz`, `H1216_pm1.npz`, `C1218_pm1.npz`, `H4868_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible ≤ 2000 ∪ on-box prior classical/SPK ships (through 4804/4036/3908/… + **1020/1116** Paley-II + **1032/1092/1152** Paley-I + **524/1048/548/1096/564/1128/600/1200** Double Paley-I + **4804** Paley-II prime power 7^4; **exclude** Miyamoto-only 4612/4516/4468/4388/4372/4196/4132/4
∪ verified H_{608}/H_{1216} (Double Paley-I) ∪ Sylvester ∪ Paley I/II (incl. prime powers), closed under Kronecker (limit 5000). Order 4868 is **Miyamoto-only** (not folded into SPK).

Quantity	Value
Known count	1047
4868 in SPK	**False** (Miyamoto-only bonus)
1216 in SPK	**True** (Double Paley-I seed)
608 in SPK	**True** (Paley-I seed)
4804 in SPK	**True** (Paley-II prime power 7^4)

Quantity	Value
1200 in SPK	**True** (prior Double Paley-I)
Next constructive gap	**2052** (OPEN)

5. Gap 2052 (method triage / non-finding ONLY)

Paley-I, Paley-II, Kronecker, and Miyamoto-with-Paley-conference paths for order 2052 all **FAIL** on-box arithmetic. This is a **non-finding**, not a counterexample. Gap **2052 remains OPEN**. Conjecture **OPEN**.

6. Non-claims

- No full Hadamard conjecture claim.
- `clay_closed = false` forever in this ship.
- GATE-04 collection **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / finished-ToE claims.
- Order 4868 is a **Miyamoto-only bonus** (NOT in SPK).

7. Next optional

- q=1229 → n=4916 via PaleyII(q=613)
- q=1237 → n=4948 via PaleyII(q=617)
- method-attack SPK gap **2052**
- Track B privacy / Object 1 meter await

8. Reproducibility

```
cd /workspace/experiments/band1/hadamard_4868_miyamoto
python3 verify_hadamard_4868.py
```

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `*.npz`, this PAPER.